

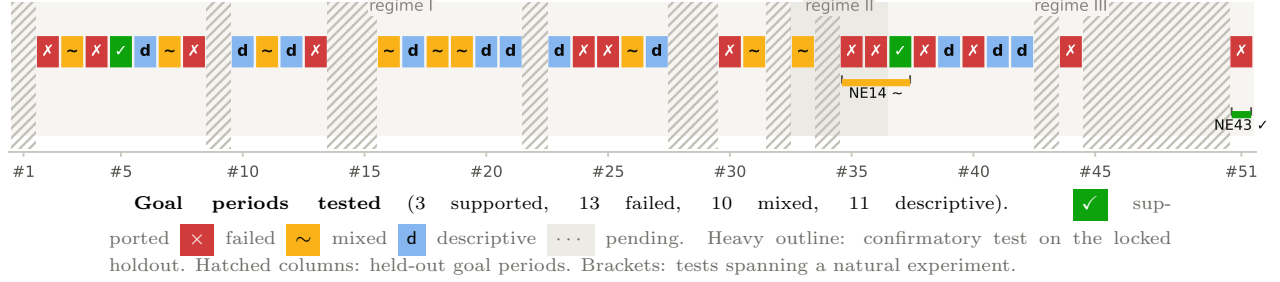
H03: Swarm activity is self-exciting, and its criticality is set by the goal's coupling mode

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. The swarm is subcritical ($n \approx 0.2$ – 0.4), its self-excitation does not depend on the goal's coupling mode, and only a small, fast cross-triggering survives its null.

level: hypothesis · pipeline checks 71% · updated 2026-10-04



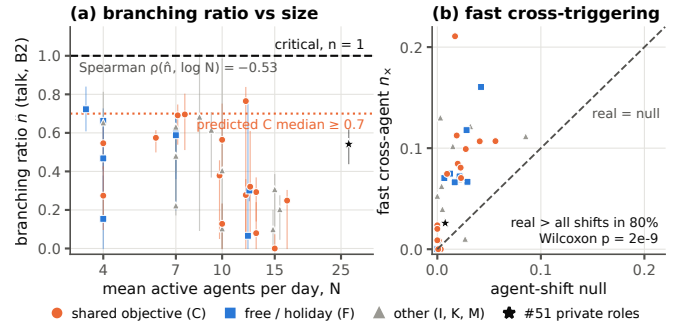
WHAT HAPPENED

The events are agent talk messages. A Hawkes fit with a schedule baseline runs in each of 35 non-holdout goal periods (282 days). The predictions are near-critical shared-objective weeks (median $\hat{n} \geq 0.7$) above free weeks, and a drift of #51 toward $n = 1$ as its roster grows. All fail. The median $\hat{n}$ is 0.41 (0.36 in round 1b); the baseline choice brackets it at 0.2–0.4. Shared-objective weeks give 0.32 against 0.53 for free weeks ($p = 0.75$). $\hat{n}$ falls with swarm size ($\rho = -0.53$) and with #51's roster ($\rho = -0.64$). A constant baseline gives a false $\hat{n} \approx 0.99$. The positive result is fast cross-agent triggering: $n_x \approx 0.07$ at 10–30 s lags, about 4× an agent-shift null ($p = 2 \times 10^{-9}$). Humans and the nudger drive about 1.5% of events directly. **Round 1b** corrects the exogenous inputs, the day cuts and the nulls. Every period verdict stays the same, and the cross-triggering survives the corrected null. In #51 (NE43), withdrawal of all operator messages leaves $\hat{n}$ and activity unchanged. No holdout run exists.

PHYSICAL MODEL

Model 09 (Hawkes self-exciting point process). The events are agent messages, with one realization per village day. $\mu_B(t)$ is the schedule baseline: a per-day level times a 30-min shape, plus a kickoff bump. g_{exo} is the human and nudger input, and ϕ is the triggering kernel. The branching ratio n is the mean number of events that one event triggers directly; $n = 1$ is critical. $\langle s \rangle$ is the mean cascade size. In the multivariate fit, λ_a is agent a 's intensity with own and cross kernels. $\bar{m}$ counts active agents, and $w_m^\times$ is the cross weight at timescale τ_m .

$$\begin{aligned} \lambda(t) &= \mu_B(t) + \sum_{x < t} g_{\text{exo}}(t - x) + \sum_{t_j < t} \phi(t - t_j) \\ \phi(\tau) &= n\beta e^{-\beta\tau}, \quad \langle s \rangle = 1/(1 - n) \\ \lambda_a(t) &= c_a\mu(t) + \sum_{t_j \in a} \phi_{\text{self}}(t - t_j) + \sum_{t_j \notin a} \phi_\times(t - t_j) \\ n_\times &= (\bar{m} - 1) \sum_{\tau_m \leq 300s} w_m^\times \end{aligned}$$



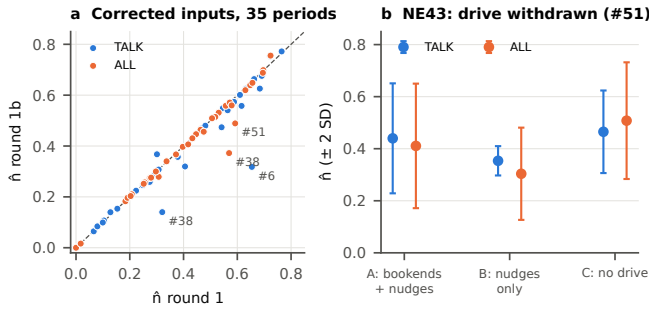
Observables. (a) The branching ratio per goal period against swarm size, by coupling mode, with day-bootstrap 95% CIs. No period is near critical. Shared-objective periods sit no higher than free ones, and $\hat{n}$ falls with size. (b) Fast cross-agent excitation against an agent-shift null that breaks cross-agent timing. Nearly every period sits above the diagonal. This component is the only one that beats its strongest null.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The swarm is far from a runaway cascade, and more agents or a shared goal do not push it closer. A monitor for message storms can assume $\hat{n} \leq 0.4$ in this scaffold: one message triggers fewer than 0.4 direct follow-ups. The scheduler and the roster set the activity level, not social contagion. The one stable coupling is about 0.07 extra messages from others within 30 s of each message. That number can be a scaffold property: agents wake when a message lands. Do not fit a constant-baseline Hawkes model to agent logs. It gives $\hat{n} \approx 0.99$ where a schedule baseline gives 0.2–0.4.

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → Verdict round 1b)	
P1 free weeks subcritical ($\hat{n}_F < 0.5$)	0.53 → 0.52; $4/8 \geq 0.5$	×
P2 shared > free	0.32 vs 0.53 → 0.29 vs 0.52 ($p = 0.80$)	×
P3 shared near critical (≥ 0.7)	0.32 → 0.29	×
P4 mode survives N , $t = -0.19$ → -0.67 ; $\log N t = -2.06$		×
P5 #51 drifts toward $n = \rho = -0.64$ → -0.66 (own × 1 split), -0.70 (shared units)		×
P6 fast $n_\times$: shared > free	0.075 vs 0.071 → 0.075 vs 0.073	×
Fast $n_\times$ vs agent-shift null	0.070 vs 0.017 → 0.074 vs 0.011 after trimming ($p = 2 \times 10^{-8}$)	✓
Native NE43: drive withdrawn	$\hat{n}$ 0.44/0.35/0.47 (within 2 SD); talk rate -11%	✓
Native #2 / NE14	$n_\times$ 0.16 vs null max 0.11; trim moves $\Delta \leq 0.03$, $\hat{n}_{\text{TALK}}$ rises	✓/∼



Second figure. (a) Per-period $\hat{n}$ in round 1 against round 1b. The corrected exogenous drive and the day cuts move almost nothing. #6 and #38 change because round 1b cuts a long operator-off gap out of one day. (b) NE43 inside #51 at a fixed roster: $\hat{n}$ does not fall when the daily bookends stop (B) or when the nudger stops (C).

GOAL PERIODS IN DETAIL

G02 × 1b: failed : n TALK 0.72 → 0.75 on corrected inputs;...
G03 ∼ 1b: mixed : n TALK 0.66 → 0.66 on corrected inputs;
...
G04 × 1b: failed : n TALK 0.55 → 0.55 on corrected inputs;...
G05 ✓ 1b: supported : n TALK 0.15 → 0.15 on corrected inputs;
...
G06 d 1b: descriptive : n TALK 0.65 → 0.32 on corrected inputs;
...
G07 ∼ 1b: mixed : n TALK 0.47 → 0.47 on corrected inputs;
...
G08 × 1b: failed : n TALK 0.27 → 0.26 on corrected inputs;...
G10 d 1b: descriptive : n TALK 0.22 → 0.22 on corrected inputs;
...
G11 ∼ 1b: mixed : n TALK 0.61 → 0.60 on corrected inputs;
...
G12 d 1b: descriptive : n TALK 0.63 → 0.63 on corrected inputs;
...
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Pooled $\hat{n}$ is still not identified against 10–30-min rate modulation. In regime III, `events_core` lacks computer-use turns, so ALL has a different meaning there. The M3 at-risk set ignores rooms. The #focus room confounds NE43's sides, and side C has 5 days. Round-1b bootstraps use 50/25 replicates (round 1: 100/50). In #51, N is collinear with date and drive. This round does not model the read-out kernel; H42 fits it.

ROUND 2

H42 owns the kernel. Next for H03: an autonomy fraction from the context ledger (H03-R1), NE10 (first nudges), and locked holdout predictions (fast $n_\times$ above the trimmed shift null; $\hat{n} < 0.8$).

- **Essence.** How autonomous is the swarm: what fraction of its activity is self-generated rather than driven by the scaffold or humans?
- **H03-R1** Autonomy fraction: classify every turn's trigger (scheduled, external message, peer message, self-continuation) from the context ledger; autonomy = peer + self share, tracked across regimes.
- **H03-R2** The swarm dies without drive: during outages and nudger-off sessions, peer-driven activity decays with a measurable lifetime.
- **H03-R3** Criticality, if anywhere, lives in content cascades (H34), not in timing.